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Interconnect for direct memory access controllers

Abstract

A computing device is provided, including a plurality of memory devices, a plurality of direct memory access (DMA) controllers, and an on-chip interconnect. The on-chip interconnect may be configured to implement control logic to convey a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device of the plurality of memory devices. The on-chip interconnect may be further configured to implement the control logic to convey a read response from the source memory device to the primary DMA controller and one or more secondary DMA controllers of the plurality of DMA controllers.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/182,556, filed Mar. 13, 2023, which is a continuation of U.S. patent application Ser. No. 17/086,010, filed Oct. 30, 2020, now granted as U.S. Pat. No. 11,604,748, the entirety of each of which is hereby incorporated herein by reference for all purposes.

BACKGROUND

(1) System-on-a-chip (SOC) architectures have become increasingly common as demand for high-performance computing has grown. In an SOC architecture, data processing and storage capabilities are provided together in an integrated circuit rather than being provided in separate integrated circuits connected to the motherboard of a computing device. By combining processing and memory functionality into an SOC, computations may be performed more quickly as a result of not having to transfer data through the motherboard. In addition, using an SOC may allow the size and power consumption of the computing device to be reduced.

SUMMARY

(2) According to one aspect of the present disclosure, a computing device is provided, including a plurality of memory devices, a plurality of direct memory access (DMA) controllers, and an on-chip interconnect. The on-chip interconnect may be configured to implement control logic to convey a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device of the plurality of memory devices. The on-chip interconnect may be further configured to implement the control logic to convey a read response from the source memory device to the primary DMA controller and one or more secondary DMA controllers of the plurality of DMA controllers.

(3) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 schematically shows an example computing system including a system on a chip (SOC), according to one example embodiment.
- (2) FIG. 2 schematically shows the SOC of FIG. 1 when a read request and a plurality of read responses are transmitted over an on-chip interconnect.
- (3) FIG. 3 schematically shows the SOC of FIG. 1 when a plurality of read receipts are respectively conveyed to a first subset of a plurality of direct memory access (DMA) controllers but not to a second subset of the plurality of DMA controllers.
- (4) FIG. 4 schematically shows a read request and a read response, according to the embodiment of FIG. 1.
- (5) FIG. 5 schematically shows a first read response and a second read response received at a DMA controller, according to the embodiment of FIG. 1.
- (6) FIG. 6 schematically shows a first read response including a first pointer address and a second read response including a second pointer address, according to the embodiment of FIG. 1.
- (7) FIG. 7A shows a flowchart of a method for use with a computing device, according to the embodiment of FIG. 1.
- (8) FIG. 7B-7D show additional steps of the method of FIG. 7A that may be performed in some examples.
- (9) FIG. 8 shows a schematic view of an example computing environment in which the computing system of FIG. 1 may be enacted.

DETAILED DESCRIPTION

- (10) FIG. 1 shows an example computing system 1 in which a computing device in the form of an SOC 10 may be included. In addition to the SOC 10, the computing system may include memory 2, one or more processors 4, one or more input devices 6, one or more output devices 8, and/or one or more other components. The components of the computing system 1 may, for example, be electrically coupled over a motherboard. In addition, the computing system 1 may be communicatively coupled to one or more other computing systems via the one or more input devices 6 and/or the one or more output devices 8. For example, the computing system 1 may be included in a data center in which data processing and storage are performed at a plurality of interconnected computing systems.
- (11) The SOC 10 may include a plurality of memory devices 12. For example, the SOC 10 may include one or more memory devices 12 that function as dynamic random-access memory (DRAM) and one or more memory devices 12 that function as static random-access memory (SRAM).

Memory devices **12** may include other types of random access memory, as well. In addition, the SOC **10** may include one or more processing devices **14**. Each processing device **14** of the one or more processing devices **14** may, for example, be a central processing unit (CPU), a core of a CPU, a graphics processing unit (GPU), a core of a GPU, a field-programmable gate array (FPGA), or an application-specific integrated circuit (ASIC).

(12) The plurality of memory devices **12** may be coupled to the one or more processing devices **14** by an on-chip interconnect **16** via which data may be transferred. The on-chip interconnect **16** may, for example, be a network-on-chip (NoC), a crossbar, or a ring network. In other examples, other network topologies may be used to couple the plurality of memory devices **12** to the plurality of processing devices **14**. In examples in which the on-chip interconnect **16** is an NoC, the on-chip interconnect **16** may include one or more routers configured to implement packet switching. In examples in which the on-chip interconnect **16** is a crossbar, the on-chip interconnect **16** may include, for each of the one or more processing devices **14**, a plurality of electrical traces coupling that processing device **14** to each of the plurality of memory devices **12**.

(13) The SOC **10** may further include a plurality of direct memory access (DMA) controllers **20**, which may be coupled to the plurality of memory devices **12**, the one or more processing devices **14**, and the on-chip interconnect **16**. Turning now to FIG. 2, in which the SOC **10** is shown in additional detail, the SOC **10** may include a primary DMA controller **20A** and one or more secondary DMA controllers **20B**. The plurality of DMA controllers **20** may each be configured to transfer data to and from the memory devices **12** of the SOC **10**. The DMA controllers **20** may each be configured to transfer data from a source memory device **12A** to a destination memory device **12B**, as discussed in further detail below. In addition, as shown in the example of FIG. 2, each DMA controller **20** of the plurality of DMA controllers **20** may be communicatively coupled to a respective processing device **14** of a plurality of processing devices **14** included in the SOC **10**. Thus, the plurality of DMA controllers **20** may be configured to transfer data between the plurality of memory devices **12** and the plurality of processing devices **14** as well as between one or more source memory devices **12A** and one or more destination memory devices **12B**.

(14) In existing SOC, a coherent fabric or a chip-level cache is typically used when transferring data between components of the SOC. A coherent fabric is an on-chip interconnect that is configured to implement snooping logic for the plurality of DMA controllers. The snooping logic may be implemented when a secondary DMA controller requests data from the source memory. In response to receiving a read request for data from the source memory, the coherent fabric may implement the snooping logic by querying the destination memory of the primary DMA controller to determine whether the destination memory of the primary DMA controller stores the requested data. When the destination memory of the primary DMA controller includes the requested data, the requested data may be copied into the destination memory of the secondary DMA controller via the coherent fabric. Thus, cache coherency between the respective destination memory devices of the primary DMA controller and the secondary DMA controller may be maintained. However, implementing snooping logic may be slow and computationally expensive.

(15) Existing SOC may alternatively use a chip-level cache (e.g. a level 2 or level 3 cache) coupled to the on-chip interconnect. Whenever a DMA controller receives data from the source memory, the chip-level cache may be updated to include the received data. When another DMA requests the same data from the source memory, that data may instead be read from the chip-level cache. However, updating the chip-level cache increases the computational cost of operating the SOC and requires the SOC to include an additional hardware component.

(16) In order to address the above issues with existing methods of transferring data to DMA controllers in an SOC, the on-chip interconnect **16** of the SOC **10** may be configured to implement control logic **30** as discussed below for the primary DMA controller **20A** and the one or more secondary DMA controllers **20B** to implement a read broadcast or a read multi-cast via one of several possible techniques. The control logic **30** is program logic encoded in software instructions

(e.g. firmware instructions) executed by interconnect processing hardware **18** included in the on-chip interconnect **16** to control the exchange of messages through the on-chip interconnect **16**. For example, when the on-chip interconnect **16** implements the control logic **30**, the on-chip interconnect **16** may be configured to convey a read request **32** from the primary DMA controller **20A** to the source memory device **12A**. The read request **32** may be generated at the primary DMA controller **20A** and may be a request to copy specific data stored at the source memory device **12A** to one or more destination memory devices **12B** included in the SOC **10**.

(17) In some examples, a secondary DMA controller **20B** of the one or more secondary DMA controllers **20B** may be configured to transmit a synchronization request **36** to the primary DMA controller **20A**. The synchronization request **36** may be transmitted to the primary DMA controller **20A** via the on-chip interconnect **16**. In such examples, the on-chip interconnect **16** may be configured to implement the control logic **30** to convey the read request **32** from the primary DMA controller **20A** to the source memory device **12A** in response to the primary DMA controller **20A** receiving the synchronization request **36**. Thus, the secondary DMA controller **20B** may be configured to request synchronization between data stored in its destination memory device **12B** and the destination memory device **12B** associated with the primary DMA controller **20A**.

(18) At the source memory device **12A**, the SOC **10** may be configured to generate a read response **34** based at least in part on the read request **32**. When the on-chip interconnect **16** implements the control logic **30**, the on-chip interconnect **16** may be further configured to convey the read response **34** from the source memory device **12A** to the primary DMA controller **20A** and one or more secondary DMA controllers **20B** of the plurality of DMA controllers **20**. In some examples, the on-chip interconnect **16** may be configured to convey the read response **34** to each DMA controller **20** of the plurality of DMA controllers **20** included in the SOC **10**. Alternatively, as shown in FIG. 3, the on-chip interconnect **16** may be configured to convey the read response **34** to a first subset **40** of the plurality of DMA controllers **20** and not convey the read response **34** to a second subset **42** of the plurality of DMA controllers **20**.

(19) As discussed above, the plurality of memory devices **12** may further include a plurality of destination memory devices **12B** respectively coupled to the plurality of DMA controllers. Subsequently to receiving the read response **34**, the primary DMA controller **20A** and the one or more secondary DMA controllers **20B** may each be configured to write data included in the read response **34** to their corresponding destination memory devices **12B**. In this way, the data that has been read from the source memory device **12A** is not only communicated to the primary DMA controller **20A** that requested the data, but also to other DMA controllers **20** using a read broadcast or multi-cast approach. Therefore, each of the destination memory devices **12B** may keep a coordinated and up-to-date memory cache without requiring a separate chip level cache or snooping logic.

(20) FIG. 4 schematically shows information that may be included in the read request **32** and the read response **34**. In the example of FIG. 4, the on-chip interconnect **16** is shown when a read request **32** is transmitted from the primary DMA controller **20A** to the source memory device **12A** and when a read response **34** is transmitted from the source memory device **12A** to the primary DMA controller **20A**. As shown in FIG. 4, the read request **32** may include read request header **50**, which may include a sequential request indicator **52** and a response recipient indicator **54**. The sequential request indicator **52**, may, for example, be a timestamp or a sequentially assigned identification number for the read request **32**. The response recipient indicator **54** may indicate the first subset **40** of the plurality of DMA controllers **20** to which the read response **34** is configured to be conveyed. For example, the response recipient indicator **54** may include a corresponding bit for each DMA controller **20** included in the SOC **10** that indicates whether that DMA controller **20** is configured to receive the read response **34**. Thus, the sequential request indicator **52** and the response recipient indicator **54** may form response multicast metadata that identifies both the read request **32** and the plurality of DMA controllers **20** that are configured to receive a read response

34. In addition to the read request header **50**, the read request **32** may further include a source memory access location **56**, which may be a pointer to a location in the source memory device **12A**.

(21) The example read response **34** shown in FIG. **4** may include a read response header **60** and a read response payload **66**. The read response header **60** may include metadata for the read response **34**, and the read response payload **66** may include the data received from the source memory device **12A** in response to the read request **32**. The read response header **60** may include a sequential response indicator **62**, which may, for example, be a timestamp or a sequentially assigned identification number. In addition, the read response header **60** may include a pointer address **64** indicating a location in the destination memory device **12B** to which the DMA controller **20** that receives the read response **34** may be configured to write the read response payload **66**. In some examples, the read response header **60** of the read response **34** may further include a primary DMA identifier **65** that indicates which DMA controller **20** of the plurality of DMA controllers **20** is the primary DMA controller **20A** that initiated the read request **32** with which the read response **34** is associated. The read response **34** may be identified as being associated with the read request **32** by, in some examples, including the sequential request indicator **52** as well as the sequential response indicator **62** in the read response header **60**. Alternatively, the state of the read request **32** and the read response **34** may be monitored at an agent executed at the processing device **14** of the primary DMA controller **20A**.

(22) As shown in the example of FIG. **5**, at least one DMA controller **20** of the plurality of DMA controllers **20** may be configured to receive a plurality of read responses **34**. In the example of FIG. **5**, a DMA controller **20** receives a first read response **34A** at a time $t_{sub.1}$ and a second read response **34B** at a time $t_{sub.1}$. The first read response **34A** has a first sequential response indicator **62A** included in a first read response header **60A**, and further includes a first primary DMA identifier **65A** and a first read response payload **66A**. The second read response **34B** has a second sequential response indicator **62B** included in a second read response header **60B**, and further includes a second primary DMA identifier **65B** and a second read response payload **66B**. In some examples, the on-chip interconnect **16** may be configured to transmit a plurality of read responses to the primary DMA controller **20A** and the one or more secondary DMA controllers **20B** in some order other than the order in which the corresponding plurality of read requests **32** were received. In order for the respective read response payloads **66** of the plurality of read responses **34** to be written to the destination memory device **12B** in the correct order, the on-chip interconnect **16** may be configured to determine a write order **68** indicated by the respective sequential response indicators **62** of the plurality of read responses **34**. The DMA controller **20** may then write the respective read response payloads **66** of the plurality of read responses **34** to the corresponding destination memory device **12B** of the DMA controller **20** as specified by the write order **68**. In the example of FIG. **5**, although the first read response **34A** is received prior to the second read response **34B**, the on-chip interconnect **16** is configured to compute a write order **68** in which the second read response payload **66B** is written to the destination memory device **12B** at time $t_{sub.2}$ before the first read response payload **66A** is written to the destination memory device **12B** at time $t_{sub.3}$. For example, when the sequential response indicators **62** of the plurality of read responses **34** are sequentially ordered numbers, the on-chip interconnect **16** may compute the write order **68** such that the plurality of read responses **34** are written to the destination memory device **12B** in ascending sequential response indicator order.

(23) In some examples, the plurality of DMA controllers **20** may be configured to write the data included in the read response **34** to a plurality of different respective pointer addresses **64** at their corresponding destination memory devices **12B**. As shown in the example of FIG. **6**, the primary DMA controller **20A** is configured to receive a first read response **134A** including a first pointer address **164A**. Similarly, the secondary DMA controller **20B** is configured to receive a second read response **134B** including a second pointer address **164B**. Subsequently to receiving the first read

response **134A**, the primary DMA controller **20A** may be further configured to write the read response payload **66** included in the first read response **134A** to the first pointer address **164A** of a first destination memory device **112A** coupled to the primary DMA controller **20A**. In addition, subsequently to receiving the second read response **134B**, the secondary DMA controller **20B** may be further configured to write the read response payload **66** included in the second read response **134B** to the second pointer address **164B** of a second destination memory device **112B** coupled to the secondary DMA controller **20B**. As a result of writing the read response payload **66** to different pointer addresses in the first destination memory device **112A** and the second destination memory device **112B**, different memory allocation schemes may be used at the first destination memory device **112A** and the second destination memory device **112B**. Thus, more efficient memory allocation may be achieved.

(24) As shown in FIG. **6**, the primary DMA controller **20A** may be coupled to a first processing device **114A**, and the secondary DMA controller **20B** may be coupled to a second processing device **114B**. In some examples, when the primary DMA controller **20A** and the secondary DMA controller **20B** respectively receive the first read response **134A** and the second read response **134B**, the primary DMA controller **20A** and the secondary DMA controller **20B** may be further configured to load the read response payload **66** into the first processing device **114A** and the second processing device **114B** respectively. In such examples, the first read response **134A** and the second read response **134B** may respectively include first processing setting metadata **138A** and second processing setting metadata **138B**. The first processing setting metadata **138A** and the second processing setting metadata **138B** may include settings with which the first processing device **114A** and the second processing device **114B** may be configured to perform one or more computations on read response payload **66**. In some examples, the first processing setting metadata **138A** and the second processing setting metadata **138B** may indicate one or more respective preprocessing steps that may be performed on the read response payload prior to writing the read response payload to the first destination memory device **112A** and the second destination memory device **112B**. The one or more preprocessing steps may, for example, be one or more steps of a compression operation or an encryption operation.

(25) FIG. **7A** shows a flowchart of an example method **200** for use with a computing device. The computing device at which the method **200** is performed may be an SOC, such as the SOC **10** of FIG. **1**. At step **202**, the method **200** may include implementing control logic at an on-chip interconnect for a plurality of DMA controllers. The on-chip interconnect may, for example, be an NoC, a crossbar, or a ring network. The plurality of DMA controllers may be configured to perform direct memory access for a plurality of memory devices included in the SOC. The plurality of memory devices may include a source memory device, which may be an SRAM device, and a plurality of destination memory devices, which may be a plurality of DRAM devices respectively coupled to the plurality of DMA controllers. In addition, the plurality of DMA controllers may be respectively coupled to a plurality of processing devices. Each processing device of the plurality of processing devices may, for example, be a CPU, a core of a CPU, a GPU, a core of a GPU, an FPGA, or an ASIC.

(26) Step **202** may include, at step **204**, conveying a read request from a primary DMA controller of the plurality of DMA controllers to the source memory device of the plurality of memory devices. The read request may be generated at the primary DMA and may be a request to transmit data stored at the source memory device to a plurality of destination memory devices via the on-chip interconnect and two or more of the plurality of DMAs.

(27) At step **206**, step **202** may further include conveying a read response from the source memory device to the primary DMA controller and one or more secondary DMA controllers of the plurality of DMA controllers. FIG. **7B** shows additional steps of the method **200** that may be performed when performing step **206**. As shown in FIG. **7B** at step **208**, conveying the read response to the primary DMA controller and the one or more secondary DMA controllers at step **206** may, in some

examples, include conveying the read response to each DMA controller of the plurality of DMA controllers. Alternatively, as shown at step **210** of FIG. 7B, step **206** may include conveying the read response to a first subset of the plurality of DMA controllers. When step **210** is performed, step **206** may further include, at step **212**, not conveying the read response to a second subset of the plurality of DMA controllers. In some examples, the read request may include an indication of which DMA controllers of the plurality of DMA controllers are configured to receive copies of the read response.

(28) FIG. 7C shows additional steps of the method **200** that may be performed in some examples. At step **214**, the method **200** may further include transmitting a synchronization request from a secondary DMA controller of the one or more secondary DMA controllers to the primary DMA controller. The synchronization request may be transmitted via the on-chip interconnect when implementing the control logic at step **202**. Step **214** may be performed prior to transmitting the read request to the source memory at step **204**. In examples in which step **214** is performed, the method **200** may further include, at step **216**, conveying the read request from the primary DMA controller to the source memory device in response to the primary DMA controller receiving the synchronization request. Thus, when the primary DMA controller receives the synchronization request, the primary DMA controller may convey a read request to the source memory to synchronize data stored in the destination memory of the primary DMA controller and the destination memory of the secondary DMA controller.

(29) FIG. 7D also shows additional steps of the method **200** that may be performed in some examples. At step **218**, the method **200** may include receiving a plurality of read responses at a DMA controller of the plurality of DMA controllers. For example, the DMA controller may be a secondary DMA controller that receives a plurality of read responses associated with read requests it did not originate.

(30) At step **220**, the method **200** may further include, at each of the primary DMA controller and the one or more secondary DMA controllers, writing data included in the read response to their corresponding destination memory devices. In some examples, the read response may include a sequential response indicator. The sequential response indicator may be included in a read response header, and may, for example, be a timestamp or a sequentially assigned number.

(31) In examples in which step **218** is performed and the read response includes a sequential response indicator, step **220** may include, at step **222**, writing the respective data included in the plurality of read responses to the corresponding destination memory device of the at least one DMA controller in a write order indicated by the respective sequential response indicators of the plurality of read responses. For example, the write order may be set as a temporal order of respective timestamps included in the plurality of read responses. If the read responses are received at the DMA controller out of temporal order, the read responses may still be written to the destination memory device of the DMA controller in the temporal order in which the read responses were generated.

(32) In some examples, step **220** may further include step **224**. Step **224** may include, at the plurality of DMA controllers, writing the data included in the read response to a plurality of different respective pointer addresses at their corresponding destination memory devices. The pointer address for each read response may, for example, be indicated in a header of the read response.

(33) Using the systems and methods discussed above, data stored in the source memory may be shared with the plurality of DMA controllers without using large amounts of source memory read bandwidth. The above systems and methods may also allow the additional hardware complexity and computational costs associated with a coherent fabric or a chip-level cache to be avoided. In addition, since different processing settings or pointer addresses may be used for read responses received at different DMA controllers, the systems and methods discussed above may further allow for increased flexibility in the processing and storage of data included in the read responses

received at the DMA controllers.

(34) In some embodiments, the methods and processes described herein may be tied to a computing system of one or more computing devices. In particular, such methods and processes may be implemented as a computer-application program or service, an application-programming interface (API), a library, and/or other computer-program product.

(35) FIG. 8 schematically shows a non-limiting embodiment of a computing system **300** that can enact one or more of the methods and processes described above. Computing system **300** is shown in simplified form. Computing system **300** may embody the computing system **1** described above and illustrated in FIG. 1. Computing system **300** may take the form of one or more personal computers, server computers, tablet computers, home-entertainment computers, network computing devices, gaming devices, mobile computing devices, mobile communication devices (e.g., smart phone), and/or other computing devices, and wearable computing devices such as smart wristwatches and head mounted augmented reality devices.

(36) Computing system **300** includes a logic processor **302** volatile memory **304**, and a non-volatile storage device **306**. Computing system **300** may optionally include a display subsystem **308**, input subsystem **310**, communication subsystem **312**, and/or other components not shown in FIG. 8.

(37) Logic processor **302** includes one or more physical devices configured to execute instructions. For example, the logic processor may be configured to execute instructions that are part of one or more applications, programs, routines, libraries, objects, components, data structures, or other logical constructs. Such instructions may be implemented to perform a task, implement a data type, transform the state of one or more components, achieve a technical effect, or otherwise arrive at a desired result.

(38) The logic processor may include one or more physical processors (hardware) configured to execute software instructions. Additionally or alternatively, the logic processor may include one or more hardware logic circuits or firmware devices configured to execute hardware-implemented logic or firmware instructions. Processors of the logic processor **302** may be single-core or multi-core, and the instructions executed thereon may be configured for sequential, parallel, and/or distributed processing. Individual components of the logic processor optionally may be distributed among two or more separate devices, which may be remotely located and/or configured for coordinated processing. Aspects of the logic processor may be virtualized and executed by remotely accessible, networked computing devices configured in a cloud-computing configuration. In such a case, these virtualized aspects are run on different physical logic processors of various different machines, it will be understood.

(39) Non-volatile storage device **306** includes one or more physical devices configured to hold instructions executable by the logic processors to implement the methods and processes described herein. When such methods and processes are implemented, the state of non-volatile storage device **306** may be transformed—e.g., to hold different data.

(40) Non-volatile storage device **306** may include physical devices that are removable and/or built-in. Non-volatile storage device **306** may include optical memory (e.g., CD, DVD, HD-DVD, Blu-Ray Disc, etc.), semiconductor memory (e.g., ROM, EPROM, EEPROM, FLASH memory, etc.), and/or magnetic memory (e.g., hard-disk drive, floppy-disk drive, tape drive, MRAM, etc.), or other mass storage device technology. Non-volatile storage device **306** may include nonvolatile, dynamic, static, read/write, read-only, sequential-access, location-addressable, file-addressable, and/or content-addressable devices. It will be appreciated that non-volatile storage device **306** is configured to hold instructions even when power is cut to the non-volatile storage device **306**.

(41) Volatile memory **304** may include physical devices that include random access memory. Volatile memory **304** is typically utilized by logic processor **302** to temporarily store information during processing of software instructions. It will be appreciated that volatile memory **304** typically does not continue to store instructions when power is cut to the volatile memory **304**.

(42) Aspects of logic processor **302**, volatile memory **304**, and non-volatile storage device **306** may

be integrated together into one or more hardware-logic components. Such hardware-logic components may include field-programmable gate arrays (FPGAs), program- and application-specific integrated circuits (ASIC/ASICS), program- and application-specific standard products (PSSP/ASSPs), system-on-a-chip (SOC), and complex programmable logic devices (CPLDs), for example.

(43) The terms “module,” “program,” and “engine” may be used to describe an aspect of computing system **300** typically implemented in software by a processor to perform a particular function using portions of volatile memory, which function involves transformative processing that specially configures the processor to perform the function. Thus, a module, program, or engine may be instantiated via logic processor **302** executing instructions held by non-volatile storage device **306**, using portions of volatile memory **304**. It will be understood that different modules, programs, and/or engines may be instantiated from the same application, service, code block, object, library, routine, API, function, etc. Likewise, the same module, program, and/or engine may be instantiated by different applications, services, code blocks, objects, routines, APIs, functions, etc. The terms “module,” “program,” and “engine” may encompass individual or groups of executable files, data files, libraries, drivers, scripts, database records, etc.

(44) When included, display subsystem **308** may be used to present a visual representation of data held by non-volatile storage device **306**. The visual representation may take the form of a graphical user interface (GUI). As the herein described methods and processes change the data held by the non-volatile storage device, and thus transform the state of the non-volatile storage device, the state of display subsystem **308** may likewise be transformed to visually represent changes in the underlying data. Display subsystem **308** may include one or more display devices utilizing virtually any type of technology. Such display devices may be combined with logic processor **302**, volatile memory **304**, and/or non-volatile storage device **306** in a shared enclosure, or such display devices may be peripheral display devices.

(45) When included, input subsystem **310** may comprise or interface with one or more user-input devices such as a keyboard, mouse, touch screen, or game controller. In some embodiments, the input subsystem may comprise or interface with selected natural user input (NUI) componentry. Such componentry may be integrated or peripheral, and the transduction and/or processing of input actions may be handled on- or off-board. Example NUI componentry may include a microphone for speech and/or voice recognition; an infrared, color, stereoscopic, and/or depth camera for machine vision and/or gesture recognition; a head tracker, eye tracker, accelerometer, and/or gyroscope for motion detection and/or intent recognition; as well as electric-field sensing componentry for assessing brain activity; and/or any other suitable sensor.

(46) When included, communication subsystem **312** may be configured to communicatively couple various computing devices described herein with each other, and with other devices.

Communication subsystem **312** may include wired and/or wireless communication devices compatible with one or more different communication protocols. As non-limiting examples, the communication subsystem may be configured for communication via a wireless telephone network, or a wired or wireless local- or wide-area network. In some embodiments, the communication subsystem may allow computing system **300** to send and/or receive messages to and/or from other devices via a network such as the Internet.

(47) The following paragraphs describe several aspects of the present disclosure. According to one aspect of the present disclosure, a computing device is provided, including a plurality of memory devices, a plurality of direct memory access (DMA) controllers, and an on-chip interconnect. The on-chip interconnect may be configured to implement control logic to convey a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device of the plurality of memory devices. The on-chip interconnect may be further configured to implement the control logic to convey a read response from the source memory device to the primary DMA controller and one or more secondary DMA controllers of the plurality of DMA controllers.

(48) According to this aspect, a secondary DMA controller of the one or more secondary DMA controllers may be configured to transmit a synchronization request to the primary DMA controller. The on-chip interconnect may be configured to implement the control logic to convey the read request from the primary DMA controller to the source memory device in response to the primary DMA controller receiving the synchronization request.

(49) According to this aspect, the plurality of memory devices may further include a plurality of destination memory devices respectively coupled to the plurality of DMA controllers. The primary DMA controller and the one or more secondary DMA controllers may each be configured to write data included in the read response to their corresponding destination memory devices.

(50) According to this aspect, at least one DMA controller of the plurality of DMA controllers may be configured to receive a plurality of read responses. The plurality of read responses may include a respective plurality of sequential response indicators.

(51) According to this aspect, the at least one DMA controller of the plurality of DMA controllers may be configured to write the respective data included in the plurality of read responses to the corresponding destination memory device of the at least one DMA controller in a write order indicated by the respective sequential response indicators of the plurality of read responses.

(52) According to this aspect, the plurality of DMA controllers may be configured to write the data included in the read response to a plurality of different respective pointer addresses at their corresponding destination memory devices.

(53) According to this aspect, the on-chip interconnect may be configured to implement the control logic to convey the read response to each DMA controller of the plurality of DMA controllers.

(54) According to this aspect, the on-chip interconnect may be configured to implement the control logic to convey the read response to a first subset of the plurality of DMA controllers and not convey the read response to a second subset of the plurality of DMA controllers.

(55) According to this aspect, the read request may indicate the first subset of the plurality of DMA controllers to which the read response is configured to be conveyed.

(56) According to this aspect, the computing device may further include a respective plurality of processing devices communicatively coupled to the plurality of DMA controllers. Each processing device of the plurality of processing devices may be a central processing unit (CPU), a core of a CPU, a graphics processing unit (GPU), a core of a GPU, a field-programmable gate array (FPGA), or an application-specific integrated circuit (ASIC).

(57) According to this aspect, the on-chip interconnect may be a network-on-chip (NoC), a crossbar, or a ring network.

(58) According to another aspect of the present disclosure, a method for use with a computing device is provided. The method may include, at an on-chip interconnect, implementing control logic for a plurality of direct memory access (DMA) controllers. The control logic may be implemented at least in part by conveying a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device of a plurality of memory devices. The control logic may be further implemented by conveying a read response from the source memory device to the primary DMA controller and one or more secondary DMA controllers of the plurality of DMA controllers.

(59) According to this aspect, the method may further include transmitting a synchronization request from a secondary DMA controller of the one or more secondary DMA controllers to the primary DMA controller. The method may further include conveying the read request from the primary DMA controller to the source memory device in response to the primary DMA controller receiving the synchronization request.

(60) According to this aspect, the plurality of memory devices may further include a plurality of destination memory devices respectively coupled to the plurality of DMA controllers. The method may further include, at each of the primary DMA controller and the one or more secondary DMA controllers, writing data included in the read response to their corresponding destination memory

devices.

- (61) According to this aspect, the method may further include receiving a plurality of read responses at a DMA controller of the plurality of DMA controllers. The plurality of read responses may include a respective plurality of sequential response indicators.
- (62) According to this aspect, the method may further include, at the DMA controller of the plurality of DMA controllers, writing the respective data included in the plurality of read responses to the corresponding destination memory device of the at least one DMA controller in a write order indicated by the respective sequential response indicators of the plurality of read responses.
- (63) According to this aspect, the method may further include, at the plurality of DMA controllers, writing the data included in the read response to a plurality of different respective pointer addresses at their corresponding destination memory devices.
- (64) According to this aspect, implementing the control logic may further include conveying the read response to a first subset of the plurality of DMA controllers and not conveying the read response to a second subset of the plurality of DMA controllers.
- (65) According to this aspect, the read request may indicate the first subset of the plurality of DMA controllers to which the read response is configured to be conveyed.
- (66) According to another aspect of the present disclosure, a computing device is provided, including a plurality of memory devices and a plurality of direct memory access (DMA) controllers including a primary DMA controller and one or more secondary DMA controllers. A secondary DMA controller of the one or more secondary DMA controllers may be configured to transmit a synchronization request to the primary DMA controller. The primary DMA controller may be configured to transmit a read request to a source memory device of the plurality of memory devices in response to receiving the synchronization request. The computing device may further include an on-chip interconnect configured to implement control logic to convey the read request from the primary DMA controller to the source memory device. The on-chip interconnect may be further configured to implement the control logic to convey a read response from the source memory device to the primary DMA controller and the one or more secondary DMA controllers. The primary DMA controller and the one or more secondary DMA controllers may each be configured to write data included in the read response to corresponding destination memory devices of the plurality of memory devices.
- (67) It will be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated and/or described may be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. Likewise, the order of the above-described processes may be changed.
- (68) The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. An integrated circuit comprising: a plurality of direct memory access (DMA) controllers; and an on-chip interconnect configured to implement control logic to: convey a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device, wherein the read request indicates one or more secondary DMA controllers that form a proper subset of the plurality of DMA controllers; and broadcast or multi-cast a read response by conveying respective copies of the read response from the source memory device to the primary DMA controller and the one or more secondary DMA controllers, each of the primary DMA controller and one or more secondary DMA controllers having a corresponding on-chip destination memory device included in

the integrated circuit, wherein the primary DMA controller and the one or more secondary DMA controllers are each configured to write data included in the read response to the corresponding on-chip destination memory devices for the respective primary and secondary DMA controllers.

2. The integrated circuit of claim 1, wherein: a secondary DMA controller of the one or more secondary DMA controllers is configured to transmit a synchronization request to the primary DMA controller; and the on-chip interconnect is configured to implement the control logic to convey the read request from the primary DMA controller to the source memory device in response to the primary DMA controller receiving the synchronization request.
3. The integrated circuit of claim 2, wherein the on-chip interconnect is configured to synchronize respective memory caches of the on-chip destination memory devices by writing the data included in the read response to the on-chip destination memory devices.
4. The integrated circuit of claim 1, further comprising a respective plurality of processing devices communicatively coupled to the plurality of DMA controllers, wherein each processing device of the plurality of processing devices is a central processing unit (CPU), a core of a CPU, a graphics processing unit (GPU), a core of a GPU, a field-programmable gate array (FPGA), or an application-specific integrated circuit (ASIC).
5. The integrated circuit of claim 4, wherein the read response further includes processing setting metadata that indicates a preprocessing operation that the respective processing devices are configured to apply to the data included in the read response prior to writing to data included in the read response to the destination memory devices.
6. The integrated circuit of claim 5, wherein the preprocessing operation is a compression operation or an encryption operation.
7. The integrated circuit of claim 1, wherein the read request indicates the one or more secondary DMA controllers in a response recipient indicator that includes, for each of the plurality of DMA controllers, a respective bit indicating whether that DMA controller is included in the proper subset.
8. The integrated circuit of claim 7, wherein the response recipient indicator is included in a header of the read request.
9. The integrated circuit of claim 1, wherein: the read request further includes a sequential request indicator; and the sequential request indicator is a timestamp or a sequentially assigned identification number.
10. The integrated circuit of claim 1, wherein: the read request further includes a source memory access location of a read response payload included in the read response; and the data written to the on-chip destination memory devices includes the read response payload.
11. A method for use with an integrated circuit that includes a plurality of direct memory access (DMA) controllers and an on-chip interconnect, the method comprising implementing control logic at the on-chip interconnect at least in part by: conveying a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device, wherein the read request indicates one or more secondary DMA controllers that form a proper subset of the plurality of DMA controllers; broadcasting or multi-cast a read response by conveying respective copies of the read response from the source memory device to the primary DMA controller and the one or more secondary DMA controllers, each of the primary DMA controller and one or more secondary DMA controllers having a corresponding on-chip destination memory device included in the integrated circuit; and at the primary DMA controller and the one or more secondary DMA controllers, writing data included in the read response to the corresponding on-chip destination memory devices for the respective primary and secondary DMA controllers.
12. The method of claim 11, further comprising: at a secondary DMA controller of the one or more secondary DMA controllers, transmitting a synchronization request to the primary DMA controller; and at the on-chip interconnect, implementing the control logic to convey the read request from the primary DMA controller to the source memory device in response to the primary DMA controller receiving the synchronization request.

13. The method of claim 12, further comprising, at the on-chip interconnect, synchronize respective memory caches of the on-chip destination memory devices by writing the data included in the read response to the on-chip destination memory devices.
 14. The method of claim 11, wherein: the integrated circuit further includes a respective plurality of processing devices communicatively coupled to the plurality of DMA controllers; and the read response further includes processing setting metadata that indicates a preprocessing operation that the respective processing devices are configured to apply to the data included in the read response prior to writing to data included in the read response to the destination memory devices.
 15. The method of claim 14, wherein the preprocessing operation is a compression operation or an encryption operation.
 16. The method of claim 11, wherein the read request indicates the one or more secondary DMA controllers in a response recipient indicator that includes, for each of the plurality of DMA controllers, a respective bit indicating whether that DMA controller is included in the proper subset.
 17. The method of claim 16, wherein the response recipient indicator is included in a header of the read request.
 18. The method of claim 11, wherein: the read request further includes a sequential request indicator; and the sequential request indicator is a timestamp or a sequentially assigned identification number.
 19. The method of claim 11, wherein: the read request further includes a source memory access location of a read response payload included in the read response; and the data written to the on-chip destination memory devices includes the read response payload.
 20. An integrated circuit comprising: a plurality of direct memory access (DMA) controllers; and an on-chip interconnect configured to implement control logic to: convey a read request from a primary DMA controller of the plurality of DMA controllers to a source memory device, wherein the read request includes: a header including a response recipient indicator specifying one or more secondary DMA controllers that form a proper subset of the plurality of DMA controllers; and a source memory access location; and broadcast or multi-cast a read response by conveying respective copies of the read response from the source memory device to the primary DMA controller and the one or more secondary DMA controllers, each of the primary DMA controller and one or more secondary DMA controllers having a corresponding on-chip destination memory device included in the integrated circuit, wherein: the primary DMA controller and the one or more secondary DMA controllers are each configured to write a read response payload included in the read response to the corresponding on-chip destination memory devices for the respective primary and secondary DMA controllers; and the read response payload is located at the source memory access location indicated in the read request.
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